

Unit 4, Lesson 11: Interpreting and Comparing Absolute Values in the Real World

Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

Learning Target

- ☐ I can compare the absolute value of two numbers in a real-world context.

4.11.1 Warm-Up: Absolute Elevations

Death Valley, a national park in California, is at an elevation of -282 feet. The city of Tallahassee, Florida is at an elevation of 203 feet.

1. Use an inequality symbol, $<$ or $>$, to compare these two elevations.

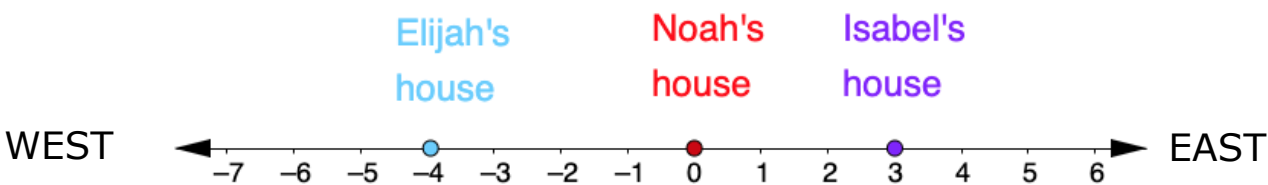
2. Use an inequality symbol, $<$ or $>$, to compare the absolute values of these two elevations.

3. Explain what these two inequality statements mean in terms of the elevations of Death Valley and Tallahassee.

4.11.2 Guided Instruction: Comparing and Interpreting Absolute Value in Context

1. Recall this scenario from a previous lesson with some additional information.

Elijah, Noah, and Isabel live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



a. Noah’s house is located at 0 on the number line model. Represent the distance from Noah’s house to both Isabel’s house and Elijah’s house using absolute value.

	Elijah’s House	Isabel’s House
Distance from Noah’s House	$ \underline{\hspace{1cm}} $	$ \underline{\hspace{1cm}} $

b. How do the absolute values in the table compare to each other?

$|\underline{\hspace{1cm}}| > |\underline{\hspace{1cm}}|$

c. Isabel lives $\underline{\hspace{1cm}}$ kilometers to the

☐ east
☐ west

 of Noah. Elijah lives $\underline{\hspace{1cm}}$ kilometers to the

☐ east
☐ west

 of Noah. Isabel lives

☐ closer to
☐ farther from

 Noah than Elijah does.

2. Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

Profit is the difference between the amount earned (revenue) and the amount spent.

His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are \$7.30, and he earns an average of \$40.00.

- a. In this situation, revenue is represented with a

☐ positive

 value and an expense is represented by a

☐ negative

 value.

- b. Write a value to represent each situation.

Landon’s Finances	Value
Expenses per lawn	
Revenue per lawn	

- c. Use an inequality symbol, < or >, to compare the absolute values of Landon’s expenses to revenue for each lawn he mows.

|____| > |____|

- d. How does the inequality show that Landon will make a profit?

- e. To make a profit, total expenses of a business must be

☐ greater than

 total revenue. A business is not profitable when the total

☐ expenses are

 greater than the

☐ revenue.

4.11.3 Collaboration: Exploring Additional Contexts

An underwater cave has an elevation of -118 feet. Here are the descriptions of four people’s elevations in relation to the cave.

- Clare’s elevation is a positive value. Clare is farther from sea level than the cave.
- The value of Andre’s elevation is smaller than the value of elevation of the cave. Andre is a farther distance from sea level than the cave.
- Kiara’s elevation is greater than the elevation of the cave. Kiara is closer to sea level than the cave.
- Lin’s elevation is the same distance away from sea level as the cave.

1. Work with your partner to complete the following table.
- Write a possible elevation for each person based on the description.
 - Use $<$, $>$, or $=$ to compare the elevation of each person to the elevation of the cave.
 - Use absolute value to compare the distance each person is from sea level to the distance the cave is from sea level.

The first row has been completed for Clare.

	Possible Elevation	Comparing Elevations	Comparison
Clare	153 feet	$153 > -118$	$ 153 > -118 $
Andre			
Kiara			
Lin			

2. Priya says her elevation is less than the cave’s and she is closer to sea level. Is this possible? Explain your reasoning or draw a picture.

4.11.4 Your Turn!

1. A bank transaction occurs when money is passed between a person and the bank. Liliana has recently made two transactions at her bank. On Monday, she deposited \$127 into her bank account. A few days later, she withdrew \$140 from her account.

- a. Write a value that could be used to represent each value in this situation.

Liliana's Bank Transaction	Value
Deposit	
Withdrawal	

- b. Use an inequality symbol, $<$ or $>$, to compare the absolute values of Liliana's transactions.

|____| ____ |____|

- c. What does this inequality statement indicate about the amount of money in Liliana's bank account after she made both transactions?

2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.

- a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin

- b. Use an inequality symbol, $<$ or $>$, to write an inequality comparing the distance each sea animal is from sea level.

|____| ____ |____|

- c. What does this inequality statement indicate about the locations of each sea animal?

4.11.5 Wrap-Up: Ticket Out the Door

1. The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	-132
Laguna del Carb3n, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	_ _ _ _ _
Fort Myers, Florida	

Britton Hill, Lakewood, Florida	_ _ _ _ _
Laguna del Carb3n, Argentina	

Laguna del Carb3n, Argentina	_ _ _ _ _
Vpadina Kaundy, Kazakhstan	

